

To study the crossover, it's important to look for the points at which the short-term average crosses the long-term average (Figure 2).

Detecting Buy and Sell crossover points

The critical difference lies in whether the price is above/below the long-term moving average or crosses the moving average.

A Buy crossover occurs when:

Yesterday: $SMA_{20} \leq SMA_{50}$

Today: $SMA_{20} > SMA_{50}$

In Python this can be implemented as:

```
buy = (
    (df["SMA20"] > df["SMA50"]) &
    (df["SMA20"].shift(1) <= df["SMA50"].shift(1))
)
```

Similarly, a Sell crossover occurs when:

Yesterday: $SMA_{20} \geq SMA_{50}$

Today: $SMA_{20} < SMA_{50}$

This can be implemented as:

```
sell = (
    (df["SMA20"] < df["SMA50"]) &
    (df["SMA20"].shift(1) >= df["SMA50"].shift(1))
)
```

The *shift(1)* is important because it allows us to compare today's relationship with yesterday's relationship.

Creating the trading position

We now convert the moving average relationship into a position by identifying the Buy/Sell signals:

```
df["Signal"] = 0
df.loc[
    df["SMA20"] > df["SMA50"],
    "Signal"
] = 1
```

Thus:

Signal = 1 → Holding the stock

Signal = 0 → Not holding the stock

We can visualise the position:

```
print(
    df[
        ["Close", "SMA20", "SMA50", "Signal"]
    ].tail(20)
)
```

Close	SMA20	SMA50	Signal
4007.268	3890.032	3879.731	1
4055.371	3908.205	3881.22	1
4157.49	3931.148	3885.056	1
4140.215	3945.39	3888.494	1
4146.408	3959.419	3891.992	1

Marking all crossovers on the chart

After knowing the position indicator, it is possible to integrate the price, the moving averages, the Buy indicators, and the Sell indicators into a visual report on the entire trading process.

```
fig, ax = plt.subplots(figsize=(15, 7))
ax.plot(
    df.index,
    df["Close"],
    label="Close Price",
    linewidth=1.2
)
ax.plot(
    df.index,
    df["SMA20"],
    label="20-Day SMA",
    linewidth=2
)
ax.plot(
    df.index,
    df["SMA50"],
    label="50-Day SMA",
    linewidth=2
)
# BUY signals
ax.scatter(
    df.index[buy],
    df.loc[buy, "Close"],
    marker="^",
    s=120,
    label="BUY",
    zorder=5
)
# SELL signals
```